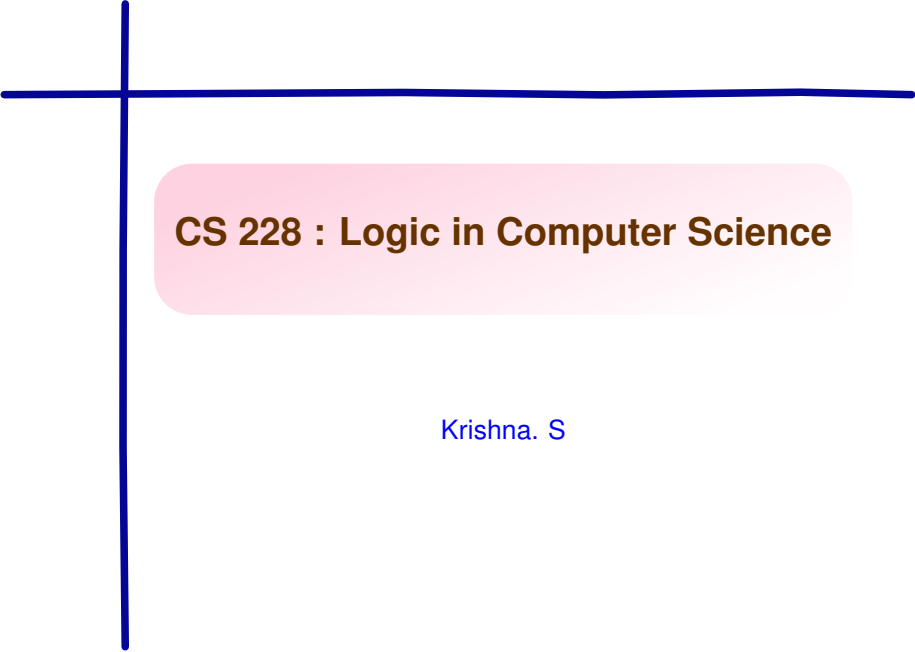




CS 228 : Logic in Computer Science

Krishna. S

Regular Languages to MSO

Given a regular language L , and a DFA such that $L = L(A)$,

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- ▶ $X_{q_0} = \{0, 2\}$, $X_{q_1} = \{1\}$, $X_{q_2} = \{3\}$
- ▶ The initial position of any word must belong to X_{q_0} : $0 \in X_{q_0}$

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Given a DFA $A = (Q, \Sigma, \delta, q_0, F)$, a word w is accepted iff it satisfies

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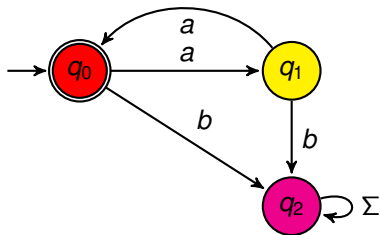
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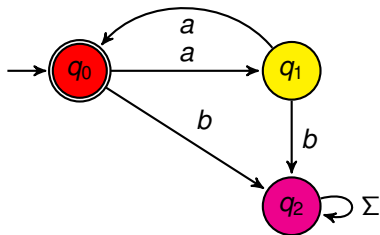
► $w \in L(A)$ iff $w \models \varphi$

Example : Regular to MSO



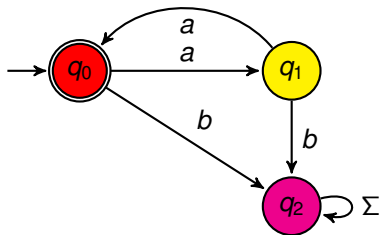
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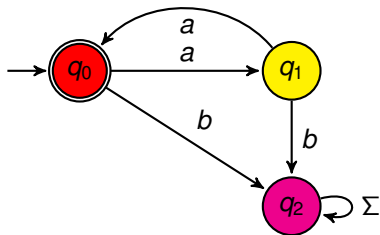
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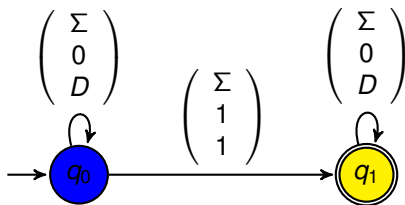
$$\wedge \exists x [\text{last}(x) \wedge (X_1(x) \wedge Q_a(x))] \}$$

MSO to Regular Languages

- ▶ Every MSO sentence φ over words can be converted into a DFA A_φ such that $L(\varphi) = L(A_\varphi)$.
- ▶ Start with atomic formulae, construct DFA for each of them.
- ▶ We already know how to handle $Q_a(x)$, $x < y$, $S(x, y)$, $x = y$
- ▶ Only $X(x)$ remains

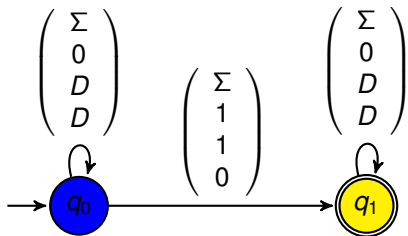
Simple Formulae to DFA

► $X(x)$



Simple Formulae to DFA

- ▶ $X(x) \wedge \neg Y(x)$
- ▶ $\Sigma' = \Sigma \times \{0, 1\} \times \{0, 1\} \times \{0, 1\}$



Formulae to DFA

- ▶ Given $\varphi(x_1, \dots, x_n, X_1, \dots, X_m)$, an MSO formula over Σ , consider the extended alphabet

$$\Sigma' = \Sigma \times \{0, 1\}^{m+n}$$

- ▶ Assign values to x_i, X_j at every position as seen in the cases of atomic formulae
- ▶ $\exists X$ handled by projecting the row of X
- ▶ If this results in non-determinism, and if you have a negation to handle, determinize and complement
- ▶ Keep in mind that every x_i should be assigned 1 at a unique position : intersect with the regular language where each of $x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n$ are assigned 1 exactly at one position

Handling Quantifiers

$\exists X \exists Y \forall x [X(x) \rightarrow Y(x)]$ On the board

The Automaton-Logic Connection

Büchi-Elgot-Trakhtenbrot Theorem (1960-1962)

Given any MSO sentence φ , one can construct a DFA A_φ such that $L(\varphi) = L(A_\varphi)$. If a language L is regular, one can construct an MSO sentence φ such that $L = L(\varphi)$.

Satisfiability of FOL

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- ▶ Undecidable in general
- ▶ Undecidability manifests with signatures containing two unary functions, one binary relation and a constant
- ▶ Decidability known on certain signatures and corresponding structures
 - ▶ FO over the word signature interpreted over word structures is one such
 - ▶ This fragment of FO is used in practical applications such as model checking where system behaviors are modeled as words

Herbrand Theory

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- ▶ Famous for Herbrand's Theorem, which allows a certain reduction from FOL to propositional logic

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- ▶ If $E(\varphi)$ is not satisfiable, then $\emptyset \in Res^*(E(\varphi))$, and we can derive this in finite number of steps
- ▶ As $E(\varphi)$ may be infinite, there is no way to say $\emptyset \notin Res^*(E(\varphi))$.

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- ▶ If the signature τ has no relations or no constants (in this case the universe is empty), there is a unique Herbrand structure for τ

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- ▶ If τ contains a constant c , function f and a unary relation R , then
 - ▶ $\mathcal{A} = (U^{\mathcal{A}} = \{c, f(c), f(f(c)), \dots\}, R^{\mathcal{A}} = \{c, f(c)\})$ is a Herbrand structure for τ .
 - ▶ $\mathcal{A} = (U^{\mathcal{A}} = \{c, f(c), f(f(c)), \dots\}, R^{\mathcal{A}} = \{c, f(c), f(f(f(f(c))))\})$ is a Herbrand structure, and so on.

Herbrand Signature

Let Γ be a set of sentences over a signature τ .

- ▶ The Herbrand signature for Γ is denoted τ_H .
- ▶ $\tau_H = \tau \cup \{c\}$ if τ contains no constants, else it is τ .
- ▶ The Herbrand universe for Γ denoted $H(\Gamma)$ is the Herbrand universe for τ_H .

Herbrand Model

A Herbrand model for Γ is a Herbrand structure M over τ_H such that $M \models \varphi$ for all $\varphi \in \Gamma$.

Satisfiability of FO

- ▶ To say that a FO sentence φ over a signature τ is satisfiable, we need to demonstrate a structure $\mathcal{A}$ over τ such that $\mathcal{A} \models \varphi$
- ▶ To say that φ is unsatisfiable, we need to say no structure $\mathcal{A}$ over τ exists so that $\mathcal{A} \models \varphi$
- ▶ What if we can say that φ is satisfiable iff it has a Herbrand structure satisfying it?

FO without equality

Let us focus on FO without “=”. Recall that “=” is always interpreted as equality.

Herbrand Theorem

Let $\Gamma = \{\varphi_1, \varphi_2, \dots\}$ be a set of equality-free sentences in Skolem Normal Form. Then Γ is satisfiable iff Γ has a Herbrand model.

If Γ has a Herbrand model, clearly Γ is satisfiable. The converse needs a proof.

The converse

Assume Γ is satisfiable. Let τ_H be the Herbrand signature for Γ .

- ▶ Let $\mathcal{A}$ be a τ_H structure such that $\mathcal{A} \models \Gamma$. ($U^{\mathcal{A}}$ need not be the Herbrand universe)
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 - ▶ Define M so that its universe is the Herbrand universe over τ_H .
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 - ▶ How to interpret the relations from τ_H (same as those of τ)?
 - ▶ Let M interpret relations like $\mathcal{A}$ (not obvious, their universes are not the same.)

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- ▶ For each n -tuple $(t_1, \dots, t_n)$ with t_i coming from the Herbrand universe $H(\Gamma)$, we must say whether $(t_1, \dots, t_n) \in R^M$ or not
- ▶ Each $t_i \in H(\Gamma)$ is a ground term in τ_H (so variable free).
- ▶ Since $\mathcal{A}$ is a structure over τ_H , if $t \in H(\Gamma)$ is a ground term from τ_H , $\mathcal{A}$ interprets t as an element of $U^{\mathcal{A}}$.
- ▶ For each n -tuple $(t_1, \dots, t_n)$, $t_i \in H(\Gamma)$, we know whether $(t_1, \dots, t_n) \in R^{\mathcal{A}}$ or not
- ▶ Define $(t_1, \dots, t_n) \in R^M$ iff $(t_1, \dots, t_n) \in R^{\mathcal{A}}$. (why does this suffice?)

Building the Herbrand Model M

- ▶ Let R be an n -ary relation in τ_H (hence in τ).
- ▶ For each n -tuple $(t_1, \dots, t_n)$ with t_i coming from the Herbrand universe $H(\Gamma)$, we must say whether $(t_1, \dots, t_n) \in R^M$ or not
- ▶ Each $t_i \in H(\Gamma)$ is a ground term in τ_H (so variable free).
- ▶ Since $\mathcal{A}$ is a structure over τ_H , if $t \in H(\Gamma)$ is a ground term from τ_H , $\mathcal{A}$ interprets t as an element of $U^{\mathcal{A}}$.
- ▶ For each n -tuple $(t_1, \dots, t_n)$, $t_i \in H(\Gamma)$, we know whether $(t_1, \dots, t_n) \in R^{\mathcal{A}}$ or not
- ▶ Define $(t_1, \dots, t_n) \in R^M$ iff $(t_1, \dots, t_n) \in R^{\mathcal{A}}$. (why does this suffice?)
- ▶ Prove that if $\mathcal{A} \models \varphi$ for any $\varphi \in \Gamma$, then $M \models \varphi$.
- ▶ The proof is by induction on the number of quantifiers in φ . Recall that each φ is in Skolem Normal Form.